

EDUGENIE — PROPOSED SOLUTION

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OVERVIEW

EduGenie is a single-page web application with five AI-powered features.

Each feature is exposed as a FastAPI route and called from the browser using JavaScript fetch requests. The backend forwards prompts to Google Gemini and returns the response.

USER FLOW

Step 1: Student opens `http://127.0.0.1:8000`

Step 2: Chooses one of five cards:

- Ask a Question
- Need an Explanation?
- Summarize a Paragraph
- Generate a Quiz
- Learning Recommendations

Step 3: Enters input and clicks the corresponding button

Step 4: JavaScript sends request to FastAPI backend

Step 5: Backend calls Gemini with a tailored prompt

Step 6: Response appears in the output box below the input

Step 7: Student can repeat any feature instantly

KEY FEATURES

Feature 1: Q&A

- Input: any academic question
- Output: concise, accurate answer

Feature 2: Explanation

- Input: topic name
- Output: simple-language explanation for students

Feature 3: Summarization

- Input: long paragraph
- Output: short, clear summary retaining key points

Feature 4: Quiz Generator

- Input: topic or passage
- Output: 3 MCQs in JSON format with 4 options each

Feature 5: Learning Path

- Input: topic (e.g., SQL)
- Output: beginner → intermediate → advanced roadmap with resources

TECHNICAL ADVANTAGES

- Zero setup for the user (no login, no downloads)
- Under 15 seconds per response
- Works on any device with a browser
- Free forever (uses free Gemini tier)